

Introduction of Simple Scalar

October 9, 2025

Contents

1	Introduction to SimpleScalar	2
2	Overview of Available Simulators	2
3	Setup and Compilation Guide	2
3.1	Step 1: Initial Setup and Navigation	3
3.2	Step 2: Compiling for the PISA Architecture	3
3.3	Step 3: Compiling for the Alpha Architecture	3

1 Introduction to SimpleScalar

SimpleScalar is an open-source computer architecture simulator that provides a framework for modeling and evaluating virtual computer systems. It allows students and researchers to test hardware designs without having to physically build them. The toolset models a complete system, including a CPU, cache, and memory hierarchy, and it can run real programs compiled for a range of modern processors.

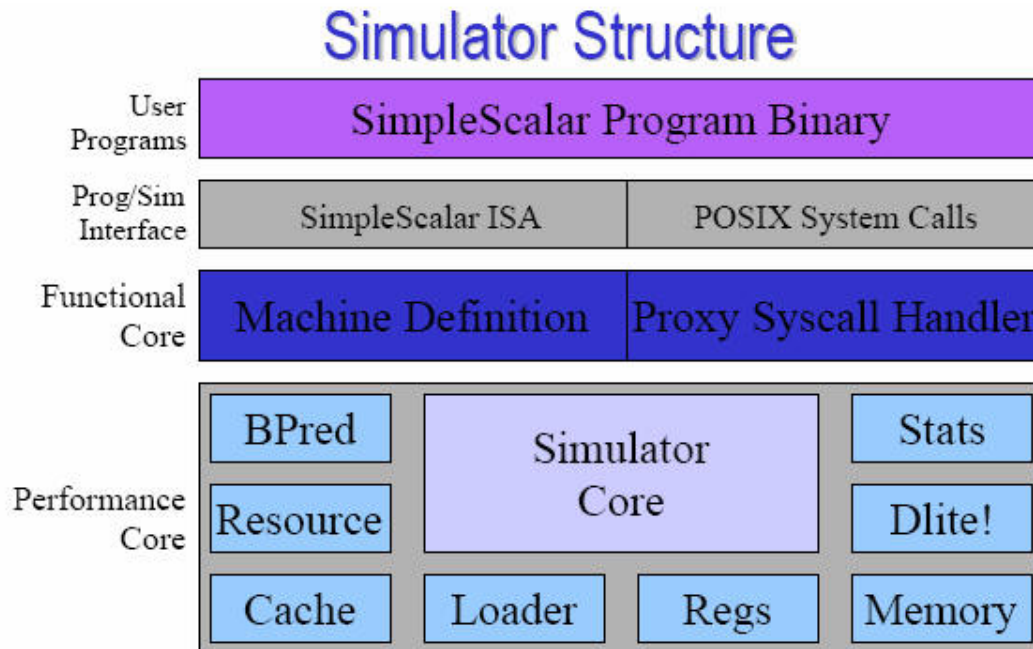


Figure 1: The architectural layers of the SimpleScalar simulator.

The simulator can target different Instruction Set Architectures (ISAs), most notably:

- **Alpha:** A real-world 64-bit RISC ISA.
- **PISA (Portable ISA):** A simpler, MIPS-like architecture designed for instructional purposes.

2 Overview of Available Simulators

The toolset includes several simulators, each offering a different level of detail.

sim-fast The fastest simulator, performing only functional simulation without modeling performance details like caches or pipelines.

sim-safe A slower version of **sim-fast** that adds memory safety checks, useful for debugging.

sim-profile A profiler that collects statistics like instruction counts and address mode usage.

sim-cache A dedicated memory system simulator for modeling multi-level caches.

sim-bpred A specialized simulator for modeling and evaluating various branch prediction schemes.

sim-outorder The most detailed performance simulator, modeling a complex, out-of-order microprocessor with caches, branch prediction, and more.

3 Setup and Compilation Guide

Before you can use the simulators, you must compile them from the source code. This process involves configuring the build system for a specific target architecture (like PISA or Alpha) and then running the compiler.

3.1 Step 1: Initial Setup and Navigation

First, ensure you are in the correct directory, as all commands depend on it.

1. **Download and Extract:** Download the `SimpleScalar.zip` file and extract its contents. This will create a folder named `SimpleScalar`.
2. **Navigate to the Source Folder:** The folder containing the source code is `simplesim-3.0`, which is located *inside* the `SimpleScalar` folder. The correct path is essential.
3. **Open a Terminal:** Open a command-line terminal window directly within the `simplesim-3.0` folder. The full path in your terminal should end with `.../SimpleScalar/simplesim-3.0`.

3.2 Step 2: Compiling for the PISA Architecture

Follow these commands in order to build the simulators for the PISA target.

1. **Clean Previous Builds** This command removes any old compiled files to ensure a fresh start.

```
1 make clean
```

2. **Configure for PISA** This step tells the build system to prepare for compiling for the PISA architecture.

```
1 make config-pisa
```

3. **Compile the Simulators** This final command starts the compilation process. When it is finished, all the simulator executables (e.g., `sim-fast`) will be available in the current folder.

```
1 make
```

3.3 Step 3: Compiling for the Alpha Architecture

To switch targets and compile for the Alpha architecture, the process is very similar. It is critical that you clean the old PISA configuration before building for Alpha.

1. **Clean Previous Builds** First, remove the existing PISA-configured files.

```
1 make clean
```

2. **Configure for Alpha** Next, configure the build system for the Alpha architecture.

```
1 make config-alpha
```

3. **Compile the Simulators** Finally, run the `make` command again to build the simulators for the Alpha target.

```
1 make
```

After completing these steps, your simulators will be ready to execute Alpha-based program binaries.